

Insulin-like growth factor-1 and childhood cancer risk



Overexpression of growth factors and/or their receptors is a common event in malignancy and provides the underlying mechanisms for one of the hallmarks of cancer, uncontrolled proliferation. Mounting evidence suggests that IGF-1 is involved in the pathogenesis and progression of different types of human cancer such as colon, breast, prostate and lung. However, only a few studies have investigated the association between IGF-1 levels and childhood cancer risk. We aimed to compare the IGF-1 serum level in children with de novo malignancies to healthy children, and to assess its relationship with cancer type, stage, metastasis and different disease characteristics. The study was carried out on 100 children; 50 children with de novo malignancies and 50 healthy children of matched age and gender as a control group. The patients were subjected to a routine work-up for their cancers according to our local standards. Estimation of the serum level of IGF-1 was carried out in the two groups using ELISA. Our results showed that children with cancer had significantly higher levels of IGF-1 than healthy controls of the same age and gender. No association was found between IGF-1 and tumor type, stage, metastasis and other disease characteristics. In conclusion, the IGF-1 serum level is an important indicator of risk for the most prevalent forms of childhood cancer. It may be used to identify children at the highest risk for these cancers and aid in determining who may benefit most from preventive strategies. Given the small number of children in our study, studies with larger populations are required to confirm these results.